



Review of methods to account for the solar spectral influence on photovoltaic device performance

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ABSTRACT

The accuracy of photovoltaic (PV) performance **forecasts** is essential for improving grid penetration, fault detection, and financing of new installations. Failing to account for the **spectral influence on PV performance** can lead to **weekly errors of up to 14% even for relatively stable technologies** such as polycrystalline silicon. There exist a range of models, known as **spectral correction functions (SCFs)**, to account for the spectral influence on PV performance **forecasts**. These SCFs use different methods to characterise both the shift in PV performance due to the spectrum, and the solar spectrum itself. This review analyses the **merits and limitations of seven commonly used spectral characterisation indices — five proxy variables (air mass, clearness index, precipitable water, aerosol, diffuse solar radiation ratio) and two variables extracted from the spectral distribution (average photon energy, depth of a water absorption band)**. The same analytical approach is adopted to review a further four indices (mismatch factor and its variants, (weighted) useful fraction, normalised short-circuit current) that are commonly used to characterise the variation in PV performance due to the solar spectrum. A review of ten SCFs that are based on these indices is undertaken to analyse the current state of the art of spectral correction modelling. The results of the review show that **whereas some proxy-variable methods offer a simple and convenient way to account for the spectral influence in PV performance forecasts, they are surpassed in terms of accuracy by SCFs based on parameters derived directly from the spectrum, such as the average photon energy and the depth of spectral absorption bands**. A decision-making framework is proposed to guide PV performance modellers in their choice of spectral correction model. The framework considers system specifications, climate, data availability, etc. The results of this work may be applied in, for example, software packages for PV performance forecasting to enable more accurate case-specific power forecasts. In future work, a standardised comparison of all SCFs and their respective indices is necessary to quantify the differences between a wider range of models than is currently available in the literature and substantiate the proposed framework.

1. Introduction

Photovoltaic performance modelling (PVPM) is crucial to PV system design, maintenance, and the integration of PV-generated energy into national energy infrastructure [1,2]. Firstly, forecasts of PV performance under specified environmental and system boundary conditions enable design optimisation prior to the investment of the significant capital that is often required in PV deployment [3]. Secondly, performance forecasts can be used as a benchmark against which measured performance can be compared in order to detect and diagnose system faults and failures in advance, which can lead to faster repair and better overall maintenance of PV systems [4]. Thirdly, timely predictions of performance help mitigate the issues caused by the intermittency of solar energy by helping energy suppliers to plan in advance the supply of energy from a variety of sources [5–8].

Models to forecast PV output must consider the influence of a variety of environmental variables such as **temperature, the incident solar spectrum, and the availability of solar irradiance** [9–15]. The impact on PV performance of the solar spectrum in particular has been discussed widely in the literature [12–14,16–21]. The performance of multicrystalline silicon (mSi) PV, which currently dominates the global PV market [22], has been reported to vary by 7%–15% with respect to standard test condition (STC) performance, depending on location, as a result of spectral variation in the field [23,24]. Other types of PV device also exhibit significant performance variation due to the spectrum. The performance of amorphous silicon (aSi) and concentrating photovoltaics (CPV) have been reported to vary by up to 20% [25–28] and 30% [29], respectively. Failing to include the effect of the spectrum in predictions of PV performance leads to forecasting errors of

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Nomenclature

α	Angstrom exponent
$\hat{\alpha}_{I_{sc}}$	Short-circuit current temperature coefficient
λ	Wavelength
Φ_{λ}	Spectral photon flux density
ρ_v	Surface water vapour density
E_{λ}	Spectral irradiance
e_s	Saturated water vapour pressure
h	Planck's constant
H_v	Apparent water vapour scale height
q	Electron charge
R_d	Diffuse solar radiation ratio
R_H	Relative humidity
T	Temperature
(W)UF	(Weighted) useful fraction
φ	Average photon energy
ϵ	Depth of a spectral band
I_{scn}	Normalised short-circuit current
K_t	Global horizontal irradiance
R^2	Coefficient of determination
$\mathbf{R}$	Earth radius vector
W	Atmospheric precipitable water content
AOD	Aerosol optical depth
AOI	Angle of incidence
DHI	Direct horizontal irradiance
e-h	Electron-hole
Exp1D	1-D exponential function
GEI	Global extraterrestrial irradiance
GHI	Global horizontal irradiance
M	Spectral mismatch factor
MAE	Mean absolute error
P	Pressure
Poly2D	Two-dimensional polynomial function
r	Pearson's correlation coefficient
RMSE	Root mean squared error
S	Solar constant (1367 W m^{-2})
z	Zenith angle

a similar magnitude. A broader review the effects of the solar spectrum on different types of PV device is presented in Ref. [30], but their work does not discuss how these spectral effects can be accounted for in PV performance models.

To account for the influence of the spectrum in PVP, a spectral correction function (SCF) is used to translate the PV output known at a set of reference conditions to the performance under arbitrary conditions. Such functions consist of two components — one component is the mathematical characterisation of the solar spectrum, while the second is the characterisation of the PV performance under the influence of variable spectral irradiance conditions. The main discussion in this paper is deconstructed in a similar manner, with a separate discussion of each of the two SCF components, and then the SCFs themselves. A select few of the published spectral and PV performance characterisation parameters have been reviewed separately in the literature [31,32], but these studies do not consider the practical applications of such parameters in the field of PVP or how these parameters can be used together in SCFs. Following an overview of a broader range of spectral and PV performance characterisation parameters, this study is the first to analyse critically how these parameters can be combined in SCFs,

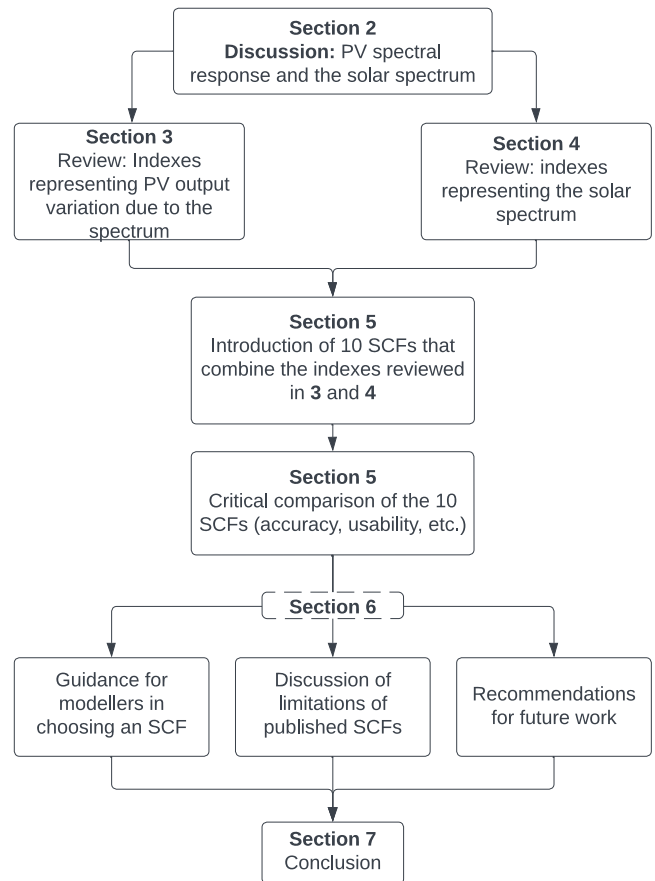


Fig. 1. The structure of this review.

quantitatively review a range of published SCFs, and offer guidance to PV performance modellers in their choice of SCF according to their system specifications, prevailing meteorological conditions, etc.

PV performance models as a whole have been reviewed extensively in the existing literature [33–37]. However, these studies do not investigate the specific spectral correction component of the overall forecasting model. In one review of PV power forecasting models, Barbieri et al. critically analyse the use of sky imaging techniques to model spectral variation caused by clouds [38]. Similarly, Antonanzas et al. mention one method of signal analysis adopted in the literature to filter solar irradiance values based on cloud-induced spectral variation [39]. However, the implications of these spectral modelling techniques for PV performance forecasting are not discussed in detail. Although Ahmed et al. do consider the implications for PVP of some methods to account for the effects of its variation on PV performance [40], a persistent shortfall of these reviews is that the scope of spectral correction methods included is narrow. In addition, there tends to be a bias towards cloud-induced spectral effects, despite there being multiple other parameters that have a significant effect on the spectrum, such as air mass, turbidities, ozone, and the variable presence of other atmospheric gases such as water vapour and SO_2 [41–45]. The importance of these additional parameters is affirmed by Duck and Fell who compare two spectral correction functions — one based on solar air mass and one based on both air mass and clearness index [46]. They conclude that an improvement in accuracy can be achieved by applying a correction for atmospheric precipitable water content to the second function so as to include information on as many spectral-related parameters as possible in the model.

The main aim of this review is to analyse 10 published methods to account for the spectral influence on PV output. Studies published over

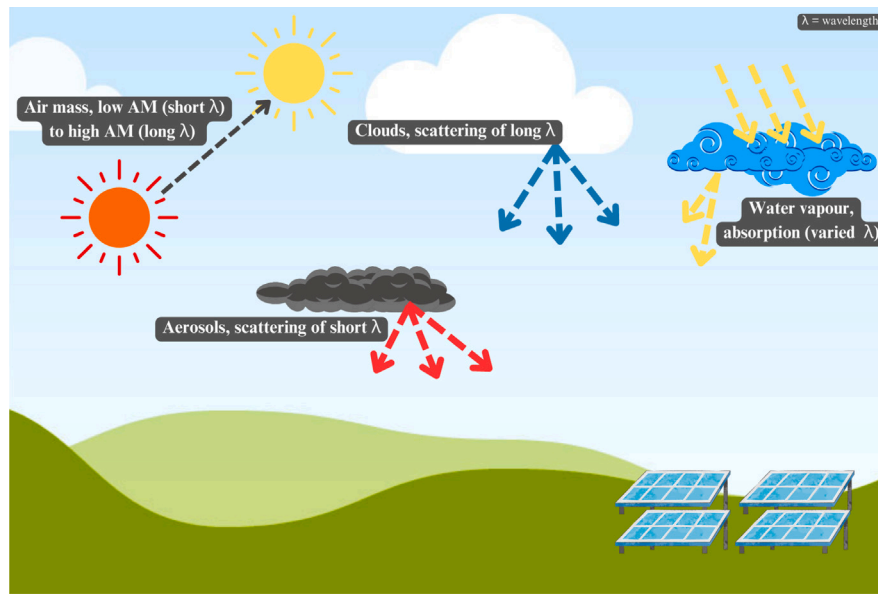


Fig. 2. Illustration of the effects of some environmental variables on spectral irradiance incident on the Earth's surface.

the last 25 years examining the relationship between PV performance and the spectrum were identified, and those **presenting a parameterisation of the relationship as an SCF** were retained. This review considers a broader scope of methods than previously considered in the published literature. In addition, the different parameters used in such methods to characterise both the solar spectrum and the variation in PV performance due to the spectrum are each individually analysed as well as the overall model performances. Based on these analyses, a decision-making framework is presented to help PV performance modellers optimise their SCF selection according to their specific use-case scenario. Furthermore, suggestions for the future direction of research in the field of spectral corrections are made.

The structure of this review is illustrated in Fig. 1. In Section 2, first, an overview of the physical phenomena underlying the spectral irradiance is presented. The environmental parameters that can influence the spectrum, such as aerosols, solar air mass, and cloud cover, and how these parameters can be used as proxies to characterise the spectrum mathematically, are also discussed. This section also explains the origins and implications of the spectral response characteristics of PV devices. In Section 3, the different methods of characterising the variation in PV performance that is solely due to variation in the solar spectrum are introduced. How the methods of spectral and PV performance characterisation are combined into a spectral correction function is discussed in Section 4, where 10 published spectral correction functions are introduced. These models are critically reviewed in Section 5, where an in-depth comparison of their accuracies and usability is presented. The usability is evaluated in terms of availability of model coefficients, extent of model validation, complexity, and so on. Based on the findings of the discussion and analysis in Sections 4 and 5, a framework is presented to guide PV performance modellers on their choice of SCF depending on their system, location, data availability, etc. Recommendations for future work in the field of spectral correction functions is presented in Section 6, while conclusions of this review are drawn in Section 7.

2. The spectrum, spectral response, and characterisation methods

In this section, an explanation of what the **solar spectrum** is and the information it contains is presented. In addition, the key factors that affect the solar spectrum, and how these may be used as proxies to characterise it mathematically, are introduced. Fig. 2 is a simple illustration of the effect of some of these environmental variables on the

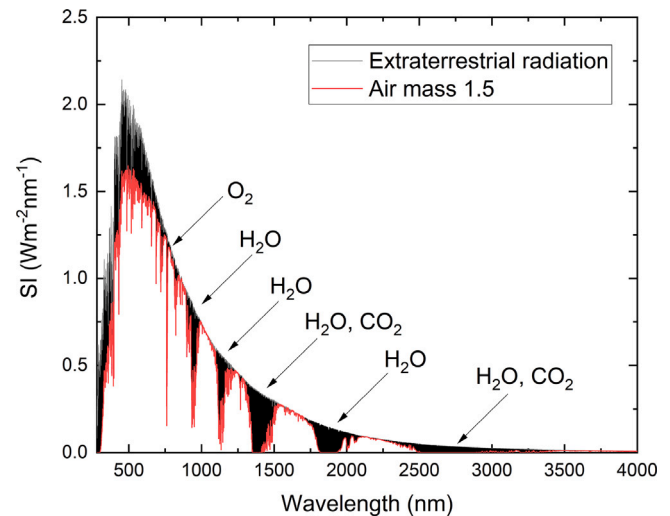


Fig. 3. Effects of atmospheric absorption and scattering on the extraterrestrial solar spectrum. The plotted data are extracted from the ASTM standard publication [50].

solar spectrum as measured on the Earth's surface. Other parameters of characterising the solar spectrum that are not based on proxies are also discussed. These variables are a key component of SCFs as they are used as the dependent variable in these functions.

2.1. Spectral irradiance and the spectral response

Spectral irradiance (SI) is defined as energy incident on a horizontal unit area per unit time per unit wavelength, in $\text{W m}^{-2} \text{nm}^{-1}$. The solar spectrum contains information on the distribution of irradiance across different wavelengths. Terrestrial spectral irradiance differs from that which is emitted by the sun due to two phenomena: (1) **Scattering** by air molecules, water, and dust; (2) **Absorption** by ozone (shorter wavelengths) and water vapour and carbon dioxide (longer wavelengths) [47–49]. The effects of atmospheric scattering and absorption are illustrated in Fig. 3, which shows the extraterrestrial solar spectrum and the reference air mass (AM) 1.5 spectrum [50].

In a photovoltaic device, it is an incident photon that provides the energy required to excite an electron from the valence to the

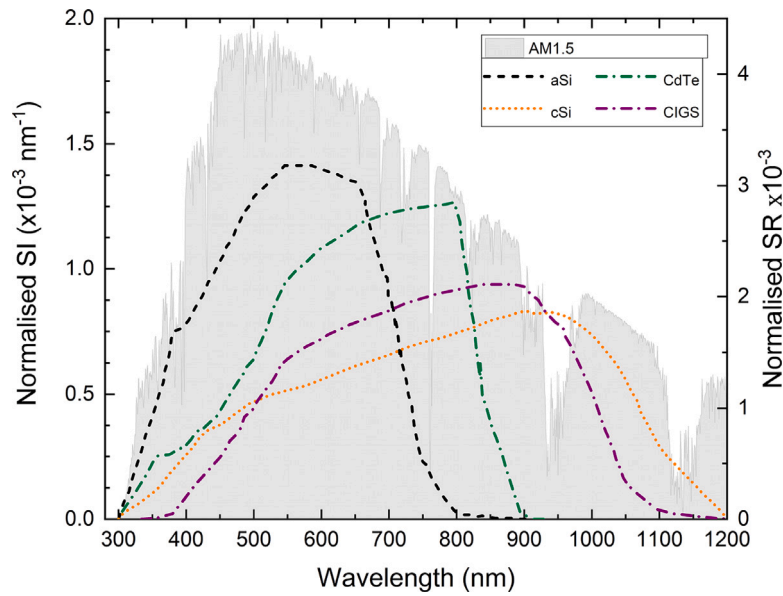


Fig. 4. Spectral responses of several common PV technologies, namely amorphous silicon (aSi), Cadmium Telluride (CdTe), crystalline silicon (cSi), and Copper Indium Gallium Selenide (CIGS). The spectral responses are plotted using data published by Amillo et al. [51]. The AM1.5 reference spectrum, normalised between 300 nm and 1200 nm, is plotted in the background for comparison.

conduction band, leaving a hole in its place. This charge separation creates an electric field that provides the force to drive the current through an external circuit. The size of the gap between the valence and conduction bands is known as the band gap, E_g , and a photon (γ) must transfer an energy (E) of $E_\gamma \geq E_g$ to generate an electron-hole (e-h) pair. This relationship between PV output and incident photon energy leads to what is known as the PV spectral responsivity, which is the response of a PV module to incident photons as a function of wavelength. The spectral responsivities of several common technologies are shown in Fig. 4. The AM1.5 reference spectrum is plotted in the background.

2.2. Spectral characterisation

The different response ranges and peaks at different wavelengths for the various PV technologies means that the way in which the different atmospheric components interact with the spectrum, the effects of which are shown in Fig. 3, can have differing impacts on the performance of different PV technologies. These different atmospheric parameters, and their effects on PV performance and the solar spectrum, are discussed in this section. Some of these parameters form the basis of published spectral correction functions, since if a particular parameter's effect on the spectrum is significant then variation in this parameter may be considered indicative of variations in the solar spectrum. Hence, this parameter may be used as a single-value proxy to represent the solar spectral distribution. The following subsections introduce some of these proxy variables, as well as two more direct methods of characterising solar spectral distributions.

2.2.1. Air mass

The **path length of direct radiation** through air is described by air mass. This path length is calculated as the ratio of direct optical path length to the path length at the zenith (vertically upwards), z [52]. This geometry is illustrated in Fig. 5.

A first order approximation of air mass is:

$$AM = \sec z. \quad (1)$$

Eq. (1) assumes that the atmosphere is a flat horizontal layer, ignoring its finite height, and hence predicts infinite air mass at the horizon. Therefore, Eq. (1) hold for angles of z up to around 70° (AM

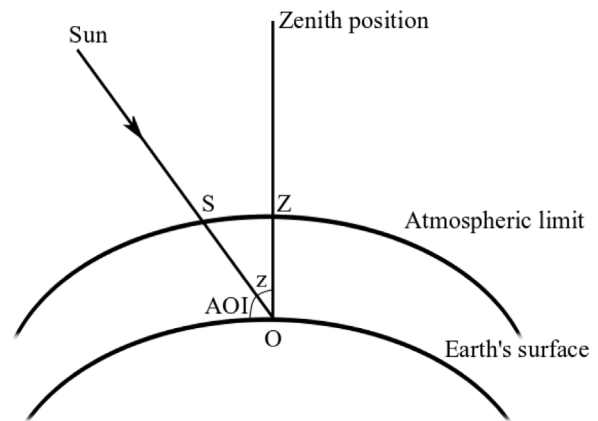


Fig. 5. Solar geometry: Angle of Incidence (AOI), Zenith Angle (z), and air mass (SO:ZO).

= 2.92) [53]. Kasten and Young propose a refined model that accounts for the curvature of the Earth [54]:

$$AM = (\cos z + 0.50572(96.07995 - z)^{-1.6364})^{-1}. \quad (2)$$

As the direct optical path length increases, greater scattering of shorter wavelength (higher frequency) light occurs, resulting in a greater proportion of longer wavelength light reaching the observer [55]. As a result, changes in air mass are strongly correlated with changes in the solar spectrum [56–58]. This relationship is shown in Fig. 6, in which the normalised spectral distributions of different air masses are plotted. The spectral irradiance curves are simulated by the authors using the SMARTS software [59,60]. The normalisation is carried out by dividing the spectral irradiance value for each wavelength by the total irradiance, which is calculated by the integral of the spectral irradiance curve with respect to wavelength.

The spectral shift associated with changing air mass has led to investigations of how air mass variation is correlated with PV performance. Passow et al. find that **variation in air mass between 1.0 and 5.0 results in performance fluctuations of up to 5% in cSi modules and over 5%**

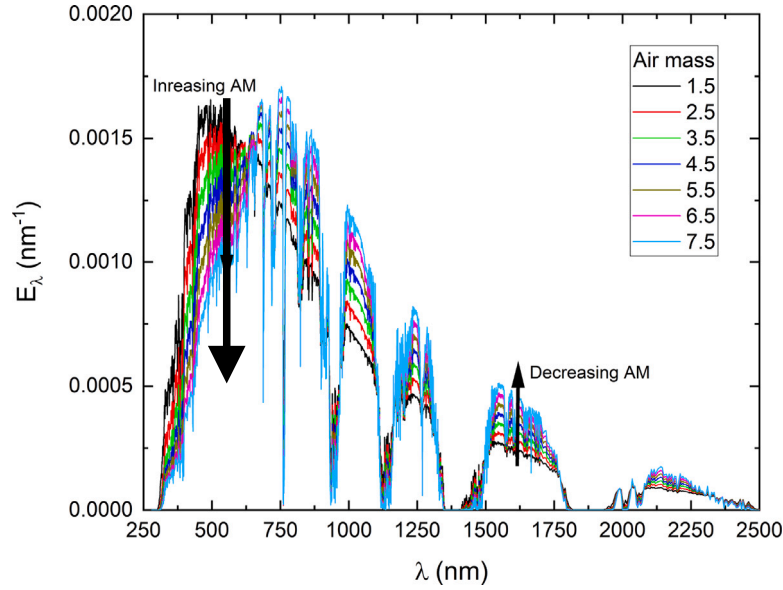


Fig. 6. SMARTS simulation of the effect of air mass on the solar spectrum.

for CdTe modules [61]. Marion et al. [62] report that, for several aSi-based technologies, changes in air mass alone can result in deviations from RTC performance of over 10%.

In certain clear sky conditions, air mass may be the dominant factor influencing the solar spectral distribution [41]. Therefore, air mass has been used in the literature to characterise the solar spectrum, since c. This correlation has been used as the basis to derive air mass-based SCFs, which are discussed in Section 4.

2.2.2. Diffuse solar radiation ratio

The diffuse solar radiation ratio, R_d , is an indicator of the level of diffuse irradiance, which is dependent on the initially available solar irradiance, atmospheric and ground conditions, and topography [63]. It is calculated simply as the ratio of horizontal diffuse irradiance to the global horizontal irradiance:

$$R_d = \frac{DHI}{GHI}, \quad (3)$$

where DHI is the diffuse horizontal irradiance and GHI is the total global horizontal irradiance. More information on the relationship between the different components of irradiance can be found in Ref. [64]. The R_d parameter has not been used widely in PV research but has been used in combination with the solar air mass to characterise the solar spectrum for PV-spectral research [65].

2.2.3. Clearness index

The clearness index, K_t , is an indicator of cloud cover. K_t is calculated as the ratio of the global extraterrestrial radiation (GEI) to the terrestrial global horizontal irradiance (GHI) [66]:

$$K_t = \frac{GHI}{GEI}. \quad (4)$$

The GEI may be calculated as follows:

$$GEI = \frac{S}{R^2} \times \cos z, \quad (5)$$

where the S is the solar constant, equal to 1367 W m^{-2} , R is the Earth radius vector, and z is the solar zenith angle.

Understanding the level of cloud cover is important when analysing the spectrum since longer wavelength irradiance tends to be scattered as it passes through clouds, resulting in a higher prevalence of shorter-wavelength irradiance reaching the Earth's surface [67–69]. An extensive review of the cloud effects on solar irradiance is provided by Calbó et al. [70].

Although clouds result in a drop in overall irradiance, and therefore lower PV output [71], cloud-induced spectral shifts can result in certain PV technologies operating more or less efficiently in cloudy conditions [72]. For example, aSi-based PV technologies were found to operate approximately 2% more efficiently in overcast conditions ($K_t = 0.2$) compared to clear sky ($K_t = 0.8$) conditions [73]. This is due to aSi having a higher spectral response at shorter wavelengths, as evident from Fig. 4. On the other hand, CdTe, mSi, and cSi PV technologies were all found to operate less efficiently in overcast conditions, in the same study. Ishii et al. [74] attribute $\pm 10\%$ performance fluctuations in hydrogenated aSi (aSi:H) PV to cloud cover. Other studies also affirm the importance of clouds on the spectrum, and hence the output of aSi [25] and cSi [75] devices in particular.

Albeit a useful characterisation of the prevailing status of the atmosphere, the clearness index suffers from some limitations. First, K_t is not independent of zenith angle and, second, for any given value of K_t within a given range of solar elevation, the atmospheric conditions can be different in terms of, for example, direct and diffuse irradiance levels [76,77].

2.2.4. Atmospheric precipitable water content

As discussed in Section 2 and illustrated in Fig. 3, water vapour content in the atmosphere is responsible for absorption of solar irradiance in specific wavebands, with a particularly strong effect in the near-infrared region [78]. One measure of water vapour content in the atmosphere is the atmospheric precipitable water vapour, W . W , usually measured in centimetres, is the height that would be reached if all of the water vapour in a column from the Earth's surface to the troposphere were to condense. A comprehensive discussion of W characteristics around the world is presented in Ref. [79] and real-time measured precipitable water data are publicly available for many locations around the world [80].

In the absence of measured data, $W[\text{cm}]$ may easily be estimated from relative humidity and temperature using the following relationship [81]:

$$W = 0.1 H_v \rho_v, \quad (6)$$

where $H_v[\text{km}]$ is the apparent water vapour scale height, $\rho_v[\text{g m}^{-3}]$ is the surface water vapour density, and the coefficient 0.1 is necessary to reconcile units. H_v is a function of temperature and is calculated as follows:

$$H_v = 0.4976 + 1.5265\beta + \exp(13.6897\beta - 14.9188\beta^3), \quad (7)$$

where $\beta = T/T_0$ and $T_0 = 273.15$ K. Meanwhile, ρ_v is a function of relative humidity (R_H), temperature, T [K], and saturated water vapour pressure, e_s [mb], which is a function of temperature [82]:

$$\rho_v = 216.7 \cdot \frac{R_H e_s}{T}, \quad (8)$$

where:

$$e_s = \exp a_0 + a_1 T_0^{-1} + a_2 T_0^{-1} + a_3 T_0^{-1}. \quad (9)$$

There exist many other models to estimate e_s , for example Refs. [83–87]. A wider range of these models is analysed extensively in Ref. [82]. The approach reported here for reference is used in [88].

Given the significant W -induced spectral irradiance absorption around the peak response of CdTe devices, a notable correlation between variation in W and CdTe performance has been highlighted in the literature. Nelson et al. [89] find a $\pm 6\%$ variation in CdTe performance due to variation between in W between 0.1 cm and 10 cm, when other variables (e.g. pressure, ozone, irradiance, etc.) are held constant.

2.2.5. Aerosol optical depth

Atmospheric aerosols are suspensions of liquid, solid, or mixed particles with highly variable chemical composition and size distribution [90]. Aerosol optical depth (AOD) is the dimensionless measure of extinction of beam (direct) irradiance caused by these aerosols [91]. Global aerosol data is available from a variety of sources [92,93].

The impact of AOD on spectral irradiance can be estimated using the Ångström turbidity formula [94]:

$$AOD = AOD_{500} \left(\frac{\lambda}{0.5} \right)^{-\alpha}, \quad (10)$$

AOD_{500} accounts for the quantity of aerosols in a column of the atmosphere at 500 nm, and λ and α are the wavelength and the Ångström exponent, respectively. AOD scattering primarily affects the UV–visible range of the solar spectrum and is negligible at wavelengths longer than about a micron [95–97], hence the impact of its variability is particularly significant for wide bandgap PV technologies such as CdTe and Perovskite [98]. AOD-induced spectral shifts have also been found to be responsible for performance changes of around 4% for concentrating triple junction PV devices [99].

Besides the impact of aerosols on the spectrum, absorption and enhanced levels of diffuse irradiance due to aerosols have also been studied more broadly in terms of their effects on PV power generation. During extreme weather events such as wildfires and smog, PV power generation has been found to be reduced by up to 40% as a result of increased AOD [100,101].

2.2.6. Average photon energy and an absorption band

Unlike the other parameters discussed thus far, the average photon energy (APE, ϕ) is not an environmental variable that affects the spectrum. Rather, the average photon energy is a value extracted directly from a solar spectral distribution that can be used to characterise it. The APE has been used widely in the Physics literature since the 1900s [102–105], but its application to characterising spectral irradiance measured on the Earth's surface, in particular for PV applications, was proposed in Ref. [106]. The APE originates from considering the inverse relationship between photon wavelength (λ) and energy (E):

$$E_\gamma = \frac{hc}{\lambda}, \quad (11)$$

where h is Planck's constant, c is the speed of light in a vacuum, and E_γ is the photon energy. If one considers the mean energy of all photons in a solar spectral distribution, this value gives an indication of the overall shape of the distribution. This mean value is calculated by dividing the total energy in the spectrum by the number of photons it contains [106]:

$$\text{APE [eV]} = \frac{1}{q} \left(\frac{\int_a^b E_\lambda d\lambda}{\int_a^b \Phi_\lambda d\lambda} \right). \quad (12)$$

Here, E_λ [W m⁻² nm⁻¹] is the spectral irradiance, Φ_λ [m⁻² nm⁻¹] is the spectral photon flux density, q [C] is the electron charge, and a [nm] and b [nm] are the upper and lower wavelength limits, respectively, of the considered waveband.

When the mean is calculated between limits a and b , theoretically there exists the possibility that increases in spectral irradiance at one wavelength may be countered by a decreases at another wavelength, thus leading to the same average photon energy but a different shape of the spectrum. If this were the case, two spectral distributions would have the same average photon energy but different shapes, and hence different effects on PV performance. In order to validate the reliability of the APE parameter to represent solar spectral distributions, Minemoto et al. adopt the IEC's methodology for rating the spectral matching of a solar simulator [107] and find that a single APE value yields a spectral irradiance distribution with a relatively small standard deviation [108]. They conclude that the APE is a bijective index that can uniquely represent different solar spectral distributions. The APE parameter has been used extensively in subsequent research to characterise the spectral irradiance distribution [109–113]. Unlike the proxy variable methods, rather than taking one variable that affects the spectrum to represent it, the APE is extracted directly from the spectrum after all possible factors have already influenced it. Therefore, it is argued widely in the literature that the APE contains information on the dominant environmental conditions influencing PV performance and is therefore capable of effectively characterising the shape of the incident spectral distribution [114–116].

Adopting the same methodology as Minemoto et al. but analysing the spectra using the coefficient of variation (CV) rather than the standard deviation, Nofuentes et al. argue that the APE is in fact not a unique characteristic of the spectral distribution [117]. They find that in the 450–900 nm waveband, the CV remains below 3.3%, which is acceptable, but outside of this waveband the CV reaches values of up to 5%–11%. Between 350–450 nm, the CV lies around 5%–6%, and reaches 6%–11% for the 900–1050 nm waveband. They conclude that the APE may only be considered approximately unique within this mid-range waveband of 450–900 nm for the climate considered in their study, but otherwise the APE is not a unique characteristic of the spectrum. They attribute the spread about the mean predominantly to experimental uncertainty for wavelengths below 450 nm, which could be mitigated in future work, but attribute it to a mixture of experimental uncertainty and the direct effects of aerosols and water vapour for wavelengths longer than 900 nm.

Much of the existing PV-APE research [118–120] tends to focus on thin-film PV devices, in particular aSi technologies. Given that the spectral response of thin-film devices is narrow, with aSi cutting off at around 750 nm to 800 nm, the issues raised by Nofuentes et al. [117] for spectral effects beyond 900 nm may not have been a problem.

Ishii et al. [121] find that the primary driver for changes in the shape of spectra that maintain the same APE are the negatively correlated depths of water absorption bands and atmospheric windows. This principle is illustrated in Fig. 7. They show that a solar spectral distribution may therefore be characterised uniquely for all wavelengths with both the APE and an additional index, namely the depth of a water absorption band, ϵ_w , or an atmospheric window, ϵ_a .

The depth of one of these spectral bands, ϵ , can be calculated by integrating the spectral irradiance with respect to wavelength within the wavelength limits of the band in question:

$$\epsilon = \int_c^d E_\lambda d\lambda. \quad (13)$$

2.2.7. Summary

The spectral characterisation indices discussed in this section are summarised in Table 1. One of the main differences between the parameters the balance they strike between simplicity and accuracy, which is determined primarily by the data requirements for their calculation. In the table, the data “level” refers to how complex, affordable,

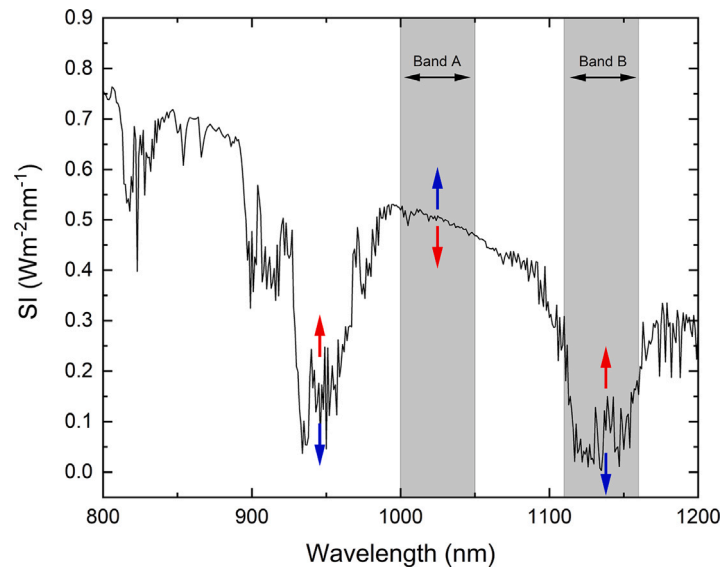


Fig. 7. Atmospheric window (Band A, 1000–1050 nm) and water absorption band (Band B, 1110–1160 nm) as highlighted in [121]. The red and blue arrows indicate a higher and lower SI in bands A and B, respectively. SI data were measured at the University of Nottingham.

Table 1

Summary of the indices used to characterise the prevailing spectral irradiance conditions. The data “level” refers to the requirement for data of different levels of accessibility or complexity. For example, irradiance data would be considered medium level due to the need for non-trivial, but still relatively simple and common measurements at any established PV or meteorological monitoring station.

Index	Eqs.	Type	+	–	Refs.
AM_d	(1), (2)	Proxy	Simple calculation, low-level data	No diffuse irradiance	[54]
R_d	(3)	Proxy	Simple calculation, medium-level data, diffuse effects included	Diffuse irradiance only best used with other parameters	[64,65]
K_t	(4)	Proxy	Cloud effects included, medium-level data	Limited information on complex scattering	[49]
W	(6)	Proxy	Accurate measure of a specific atmospheric phenomenon, simple estimation	High-level data (unless estimated), best used with other parameters	[82,88]
AOD	(10)	Proxy	Accurate measure of a specific atmospheric phenomenon	High-level data	[94]
φ	(12)	Direct	Single number with info on multiple E_λ features and atmospheric phenomena	High-level data	[73,122]
ϵ	(13)	Direct	Accurate measure of a specific atmospheric phenomenon	High-level data	[121,123]

and/or publicly accessible the required data acquisition is for the direct measurement or calculation of the parameter in question. Overall, it appears as though the characterisation of spectral effects on the PV performance through the use of proxy variables is simpler but less accurate, while the direct characterisation through the use of spectral data requires a more complex measurement but is more accurate. The simulation of spectral irradiance data to calculate parameters such as the average photon energy could mitigate the measurement issue but validation of such methods in conjunction with PV performance forecasting is not currently presented in the existing literature.

3. Characterising spectrum-induced PV performance variation

Section 2.2 describes some of the parameters that can be used to characterise an arbitrary solar spectrum. In order to derive a spectral correction function and analyse the impact of the spectrum on PV performance, it is also necessary to have a parameter that characterises the purely spectral effect on PV performance, excluding the effects of other parameters such as irradiance, temperature, etc. In this section, several such parameters are discussed.

3.1. Useful fraction and the weighted useful fraction

The useful fraction (UF) is the ratio of irradiance that falls within the spectral response of a PV device to the total available irradiance. It

was originally termed “available spectrum” by Hirata and Tani (1995) who first proposed it [27], but has since become known as the Useful Fraction. The UF is calculated as follows [17,27]:

$$UF = \frac{1}{G} \int_0^{\lambda(E_g)} E_\lambda d\lambda, \quad (14)$$

where G is the total irradiance and the interval of integration is from zero to the band gap, E_g , expressed as a wavelength (λ).

Although the UF has been used in previous publications [124–126], one of its significant limitations is that it assumes a constant (100%) spectral response across all wavelengths. An alternative is the weighted useful fraction (WUF), which weights the fraction calculated at each wavelength by the module spectral response at that wavelength [127]. The WUF is calculated as follows:

$$WUF = \frac{\int_{\lambda_{G-\min}}^{\lambda_{G-\max}} E_\lambda SR_\lambda d\lambda}{\int_{\lambda_{G-\min}}^{\lambda_{G-\max}} E_\lambda d\lambda}. \quad (15)$$

Since the true module SR is never 100% at any wavelength, the UF is always greater than the WUF. The overestimate of useful irradiance by the UF would lead to an overestimate of PV power output. One mutual limitation of the UF and WUF is that they are both device dependent and both also require knowledge of the module spectral response, which can only be acquired through a non-trivial measurement with specialist equipment.

3.2. Normalised short-circuit current

The solar spectral influence on PV performance may also be characterised through the direct use of performance measurements. The short-circuit current of a PV device, I_{sc} , is subject to the effects of many different variables, including the spectrum. However, the dominant of these are the broadband irradiance and PV cell temperature. **By isolating the non-spectral effects from the measured short-circuit current, any remaining deviation from an RTC performance can be attributed to variation in the incident solar spectrum from the reference spectrum.** Other effects considered dominant for the DUT may also be taken into account, such as the angle of incidence.

The normalisation process is as follows. I_{sc} , which is measured at arbitrary irradiance and temperature is translated to a reference temperature, T_r , and reference irradiance, G_0 , to give an intermediary current value, I'_{sc} :

$$I'_{sc} = \frac{I_{sc}}{1 + \alpha_{I_{sc}}(T_c - T_r)} \left[\frac{G_0}{G_{poa}} \right]. \quad (16)$$

$\alpha_{I_{sc}}$ [$^{\circ}\text{C}^{-1}$] is the short-circuit current temperature coefficient, T_c [$^{\circ}\text{C}$] is the measured (or modelled) cell temperature, and G_{poa} [W m^{-2}] is the measured (or modelled) plane of array irradiance. Other effects may also be removed from I_{sc} in the calculation of I'_{sc} , such as angle of incidence, for example.

The normalised short-circuit current, I_{scn} , is then found by dividing I'_{sc} by the reference short-circuit current, I_{sc0} , which is the current measured at the reference test conditions (T_r , G_0 , and $E_{\lambda,0}$):

$$I_{scn} = \frac{I'_{sc}}{I_{sc0}}. \quad (17)$$

The normalisation principle here is a standard mathematical technique and is widely known for its use in the Sandia model [41,128], and has also been adopted in other SCF literature [65,122,123,129].

I_{scn} is a device-independent parameter, thus making it easier to understand and use to compare the spectral effects on different devices. $I_{scn} > 1$ indicates higher performance under the prevailing spectrum with respect to the performance under the reference conditions, $I_{scn} < 1$ indicates decreased performance, while $I_{scn} = 1$ indicates there is no difference between the performance under the prevailing spectrum and under reference conditions.

3.3. Spectral or mismatch factor

The spectral mismatch factor (SF), or mismatch factor (M), is calculated based on the differences between the responses of the PV device under test (DUT) and a reference device, to both the incident spectral irradiance and a reference spectrum. It is calculated as follows [130]:

$$M = \frac{\int E_{\lambda,0} SR_{\lambda,0} d\lambda \int E_{\lambda,meas} SR_{\lambda,sample} d\lambda}{\int E_{\lambda,meas} SR_{\lambda,0} d\lambda \int E_{\lambda,0} SR_{\lambda,sample} d\lambda}, \quad (18)$$

where the first subscript, λ , denotes the functional dependence, and the second, *ref*, *meas*, or *sample*, indicates the reference value, measured value, or sample (DUT) value, respectively.

The calculation of M requires knowledge of the module spectral response, which, as highlighted in the discussion on UF, is a non-trivial measurement. However, the M is in fact analogous to I_{scn} [28,131]. The final output is also a ratio with values greater than, less than, or equal to unity indicating increased, decreased, and the same performance under the prevailing spectrum relative to performance under the reference spectrum. Both M and I_{scn} are ratios of performance under prevailing conditions to reference conditions, albeit calculated in different ways.

M has been used widely in the literature to understand differences between measured spectra and reference spectra, for example when calibrating solar simulators [132], testing new PV materials [133],

and for PV performance analysis under varying spectral irradiance conditions [134–136].

Variants of the spectral factor exist, with some studies using its inverse for PV-spectral analysis [115,137]. Another variant used in the literature [138,139] is the weighted average spectral factor, which is calculated by weighting the instantaneous M_i values with the broadband irradiance G_i over a period of time:

$$\langle M \rangle = \frac{\sum_i G_i M_i}{\sum_i G_i}. \quad (19)$$

However, the issue regarding the requirement for knowledge of the module spectral response, and the complexity of solving the integral at multiple time steps, persists.

3.4. Summary

Table 2 summarises the indices used to characterise the shift in PV performance resulting from shifts in the prevailing spectral irradiance conditions. There is less variety in these methods compared with the methods of characterising the solar spectral distribution. I_{scn} appears to be the best approach given that it is a device-independent parameter, unlike the UF and WUF, and it offers a similar accuracy to M without the need for spectral irradiance and spectral response (SR) measurements. However, where such measurements are available, M may be a better choice. Furthermore, I_{scn} does still require module-level performance measurements.

4. Spectral correction functions for PV performance modelling

Sections 2 and 3 have explained the parameters that can be used to represent the solar spectrum, as well as its effect on PV performance, respectively. By analysing the relationship between the two, a spectral correction function can be derived. For some combinations of these parameters, different parameterisations of the correlations have been argued in the literature to be optimal. In this section, ten spectral correction functions (arising from eight parameter combinations) are introduced and explained. Following this introduction to each of the SCFs, a critical review of these functions is presented in Section 5, where the SCFs are compared in terms of their accuracy and usability.

4.1. Air mass function

The air mass spectral correction function was first proposed by King et al. (1997) [140], and later integrated into many photovoltaic performance models such as the Sandia Array Performance Model (SAPM) [141] and the five-parameter model [142].

The air mass function uses air mass to characterise the incident solar spectrum. As an example, in the SAPM, a I_{scn} is analysed as a function of a pressure-corrected value for the air mass, commonly referred to as the “absolute air mass”, AM_a . The pressure correction is required to account for the decreased density of air molecules at different elevations [143]. The air mass is calculated using the Kasten and Young (1989) model [54], and a linear pressure correction is applied by the ratio of the site pressure, P , to the pressure at sea level, P_0 . The correction can be calculated based on measurements of on-site pressure, or estimated using:

$$\frac{P}{P_0} \approx e^{-0.0001184 \cdot h}, \quad (20)$$

where h is the altitude in metres.

The functional form of $I_{scn} = f(AM_a)$ is a fourth order polynomial, the coefficients of which are dependent on the PV technology being used. The model is as follows:

$$I_{scn} = f(AM_a) = \sum_{n=0}^4 a_n \cdot \varphi^n, \quad (21)$$

Table 2

Summary of parameters used to characterise the shift in PV performance due to spectral variation. The data “level” refers to the requirement for data of different levels of accessibility or complexity. For example, spectral response (SR) data would be considered high level due to the need for complex measurements.

Index	Eqs.	+	−	Refs.
Useful fraction (UF)	(14)	Simple calculation	Device dependent, high-level data	[27,124]
Weighted UF	(15)	Accounts for variable SR	High-level data	[127]
Normalised I_{sc} (I_{scn})	(17)	Device-independent, simple calculation, intuitive representation	Multiple measurements (various levels)	[128]
Mismatch factor (M)	(18)	High accuracy, intuitive representation	Complex calculation, high-level data (SR)	[130]
M variants	(19)	Same as above, some extra information included	Complex calculation, high-level data (SR)	[138,139]

where a_{0-4} are the PV technology-dependent coefficients. An open-access database of all of the SAPM coefficients for a variety of common PV technologies has been developed by the Sandia National Laboratory and can be found in the PVLIB Python library [144]. A comprehensive guide to determining the coefficients empirically is presented in Ref. [128].

4.2. Precipitable water content function

Due to the specific susceptibility of CdTe module performance to changes in atmospheric precipitable water content, W , Nelson et al. [89] proposed an alternative to $f(AM)$ based on W . The function was primarily targeted at, and hence validated only for, CdTe PV technology.

Using the simple model of the atmospheric radiative transfer for sunshine (SMARTS) [59], in combination with Typical Meteorological Year 3 (TMY3) files for specific locations [145] as an input, the authors first show that W is a dominant parameter when determining M for CdTe PV systems. Furthermore, a model to predict M is derived, and validated in eleven locations. The final predictive model for M is as follows:

$$M_{CdTe} \approx 0.632 + 0.134 \cdot \exp(0.976(W + 0.05)^{0.079}) \quad (22)$$

Although the results reported in Ref. [89] have been used to guide further work in the field of CdTe spectral modelling [138,146], the proposed SCF has not been widely adopted. Instead, a modified version of the model to include the effects of air mass, which is discussed in the following section, has superseded the single-variable precipitable water approach.

4.3. Air mass, precipitable water function

Although W is the primary driver for changes in M for CdTe PV modules, improvements in predictive accuracy of $M = f(W)$ have been achieved through the inclusion of an additional index. Lee and Panchula [147] proposed a two-parameter spectral correction function based on both W and AM_a . Their model is validated for both CdTe and cSi PV technologies, and shows an improvement when compared to both the W and AM_a single-variable SCFs.

The derivation of $f(AM_a, W)$ follows a similar procedure to that which is used to derive $f(W)$. The SMARTS model is used to correlate spectral variation, characterised using the spectral shift (M), with W and AM_a . The 3D surface plot is parameterised, resulting in the following predictive equation for M :

$$M = b_0 + b_1 \cdot AM_a + b_2 \cdot W + b_3 \cdot \sqrt{AM_a} + b_4 \cdot \sqrt{W} + b_5 \cdot \frac{AM_a}{\sqrt{W}} \quad (23)$$

The module-specific coefficients, b_{0-5} , for the CdTe and mSi devices investigated by Lee et al. [147] are presented in Table 3. As part of a comparison with another SCF discussed later (Section 2.2.6), Daxini et al. [123] publish the $f(AM_a, W)$ model coefficients for a triple junction amorphous silicon device. These are also included in Table 3, however, the data source and filtering conditions used to derive the coefficients are different.

Table 3

$f(AM, W)$ model coefficients for an aSi-T [123], CdTe, and mSi module [147].

	b0	b1	b2	b3	b4	b5
CdTe	0.7946	−0.05423	−0.01319	0.1724	0.08372	−0.004376
mSi	0.8409	−0.02754	−0.00792	0.1357	0.03802	−0.002122
aSi-T	0.928	−0.103	−0.0597	0.0939	0.166	0.00656

Table 4

Fitted coefficients for the Huld et al. [129] spectral correction model.

Module	mSi	CdTe
k_1	0.00172	6.43×10^{-4}
k_2	5.08×10^{-4}	1.30×10^{-4}
k_3	3.57×10^{-6}	1.08×10^{-5}

4.4. Air mass, clearness index function

One issue with the SCFs presented so far is that they do not explicitly consider the effects of cloud cover on the solar spectrum. As highlighted in the discussion in Section 2.2.3, clouds have a significant impact on the efficiency of photovoltaic devices. Gottschalg et al. [25] present a correlation between the air mass, clearness index, and the UF, hereinafter referred to as the CREST model. However, the correlation is not extended to or validated as a predictive model. In later work, Huld et al. [129] from the European Joint Research Centre (JRC) present a model based on AM_a and K_t to estimate the spectral influence on photovoltaic device performance:

$$I_{scn} = k_1(e^{-K_t} - e^{-1}) + k_2(K_t - 1) + k_3(AM - 1.5). \quad (24)$$

The JRC model coefficients, k_{1-3} , are determined through fitting to measured data. The measurement campaign took place in Ispra, Italy. The multicrystalline module data were measured over a one-year period during 2003, while the CdTe data were measured from November 2008 to June 2009. The resultant model coefficients for each module are listed in Table 4.

Another model based on air mass and clearness index is the PVSPEC model, which correlates these two variables with the spectral mismatch factor, M [23]:

$$M = a_1 K_t^{a_2} AM_a^{a_3}. \quad (25)$$

In their analysis, they use the ESRA clear sky model [148] with monthly Linke turbidity values from [149] as inputs, while air mass is calculated using the Kasten and Young model presented in Eq. (2) in Section 2.2.1 [54]. The PVSPEC model coefficients for a range of PV technologies are summarised in Table 5.

The $f(AM_a, K_t)$ approach is a popular method that has been explored widely in the literature through the examination of different parameterisations and application in different locations and for different PV technologies [49,150,151].

4.5. Air mass, aerosol, precipitable water function

The SCFs introduced thus far are either single- or double-variable functions. The dependent variables in these models are environmental

Table 5

Summary of the PVSPEC model coefficients. The FS4 modules are two different CdTe devices from the same manufacturer.

PV Technology	a1	a2	a3
mSi	0.9847	−0.05237	0.03034
mono-Si	0.9845	−0.05169	0.03034
CdTe (FS4-2)	1.002	−0.07108	0.02465
CdTe (FS4-1)	0.9981	−0.05776	0.02336
aSi	1.051	−0.1033	0.009838
CIGS	0.9791	−0.03904	0.03096

Table 6

Spectral correction model for a 3T GaInP/GaInAs/Ge CPV device based on AM_a , AOD, and W .

	Correction for:		
	AM_a	AOD	W
SF1	$ax^4 + bx^3 + cx^2 + dx + e$	$a \ln x + b$	$a \ln x + b$
SF2	$ax^4 + bx^3 + cx^2 + dx + e$	$ax^2 + bx + c$	$ax + b$
SF3	$ax^4 + bx^3 + cx^2 + dx + e$	$a \ln x + b$	$a \ln x + b$

Table 7

Spectral correction model for six PV materials based on AM_a , AOD, and W .

	Correction for:		
	AM_a	AOD	W
aSi	$ax^4 + bx^3 + cx^2 + dx + e$	$ax^2 + a_x + a$	$a \ln x + b$
Perovskite	$ax^4 + bx^3 + cx^2 + dx + e$	$ax^2 + a_x + a$	$a \ln x + b$
CdTe	$ax^4 + bx^3 + cx^2 + dx + e$	$a \ln x + b$	$a \ln x + b$
mSi	$ax^4 + bx^3 + cx^2 + dx + e$	$ax + b$	$a \ln x + b$
Mono-Si	$ax^4 + bx^3 + cx^2 + dx + e$	$ax + b$	$a \ln x + b$
CIGS	$ax^4 + bx^3 + cx^2 + dx + e$	$ax + b$	$a \ln x + b$

proxies for the spectrum. Given the many environmental variables that affect the spectrum, naturally these functions are likely to exclude the effects of certain environmental parameters. This can be problematic when applying such SCFs to different climate regions around the world. In different climate regions, the relative significance of the impact of different environmental parameters on the spectrum may vary.

Theristis et al. [152] incorporate a broader scope of spectral variables by building an SCF based on air mass, aerosol optical depth, and atmospheric precipitable water content. They propose a set of analytical equations to account for the spectral influence on each subcell of a triple-junction GaInP/GaInAs/Ge concentrating photovoltaic (CPV) system. The proposed spectral corrections for each subcell are presented in Table 6.

Caballero et al. [153] expand on the CPV spectral correction model by developing a similar SCF, based on the same dependent variables of AOD, W , and AM_a , for six single-junction PV devices. By expressing the effects of AOD and W on PV performance as different functions of AM_a , a set of spectral corrects are proposed for the six PV material types (Table 7).

For each of the six PV materials, the R^2 values for $f(AOD)$ and $f(W)$ are greater than 0.97 and 0.98, respectively. The model was validated through an outdoor experiment campaign in Jaén, Spain, from March 2016 to February 2017, inclusive. Correlations between the predicted and modelled mismatch factor yielded R^2 values from 0.87 (mSi) to 0.92 (aSi), indicating a high prediction accuracy.

4.6. Air mass cumulative distribution function

Although the triple-variable method introduced in Section 4.5 is capable of including the spectral effects of more environmental factors in PV performance modelling, the method is relatively complex. In order to maintain a high level of inclusion of factors, but reduce the number of variables to retain the simplicity of the other methods, Peng et al. [65] propose a modified air mass function. They argue that the effects of aerosol, precipitable water vapour content, cloud cover, etc.

Table 8

APE spectral correction function (Eq. (27)) coefficients for an aSi module.

	c_0	c_1	c_2	c_3	c_4
Value	−2500	5552	−4626	1415	−238

cannot be ignored, but their effects can be summarised in a single parameter. Aerosol is known to increase the diffuse fraction of total irradiance [154–156], as do W [157] and cloud cover [158]. Therefore, Peng et al. combine an “air mass cumulative distribution function” (AMCDF) with the diffuse solar radiation ratio (R_d) to form a new method of accounting for the spectral influence on the performance of PV devices. Around one week of data were extracted from a four month measurement campaign for the final analysis. The resultant model is a piecewise function dependent on the value of R_d :

$$\begin{cases} f(AM) = y = 1697 \cdot AM^{-1.987} + 1836 & R_d \leq 0.7 \\ f_2(AM) = -8.76e - 0.16y^5 + 3.745e - 0.12y^4 - 5.942e \\ -0.09y^3 + 4.254e - 0.06y^2 - 0.001341y + 1.137 & \\ I_{sc0} = \frac{I_{sc0} \cdot f_2(AM)}{1 + \alpha_{I_{sc}}(T_c - T_0)} \cdot \frac{G}{G_0} & \\ f(R_d) = 0.8381R_d + 0.1244 & R_d > 0.7 \\ I_{sc} = \frac{I_{sc0} \cdot f(R_d)}{1 + \alpha_{I_{sc}}(T_c - T_0)} \cdot \frac{G}{G_0} & \end{cases} \quad (26)$$

This model does not appear to have been adopted for spectral correction purposes by industry or other researchers in their analysis of PV performance under the influence of the solar spectrum. In addition, the proposed model is not validated through an analysis of its predictive accuracy. As a result, the AMCDF model is not included in the comparison of SCFs in the next subsection as no suitable data are available to enable a meaningful comparison. However, the model description is still presented here for completeness of the review.

4.7. Average photon energy function

To simplify the spectral correction procedure by minimising the number of variables and functions that comprise the final model, whilst still retaining information on multiple environmental parameters, Daxini et al. [122] present an SCF based on the average photon energy. This recent approach represents a shift from the focus on proxy variables to variables derived directly from measured spectral distributions.

An amorphous silicon PV panel is used as an example to demonstrate the derivation and validation of the proposed model. The data for this demonstration are extracted from a publicly available empirical dataset developed by the National Renewable Energy Laboratory (NREL) [159]. Based on the annual variation in the average photon energy over the course of the one year, the dataset is split from January to August and from September to December. The former subset of data is used to derive the model, while the latter is used to validate the model. This split is argued to include, in both datasets, the full range of APE values that occur during one year.

The functional form of the resultant model is a fourth order polynomial correlating I_{scn} with APE:

$$I_{scn} = f(\varphi) = \sum_{n=0}^4 c_n \cdot \varphi^n \quad (27)$$

The model coefficients, c_{0-1} , for the aSi device are summarised in Table 8. $f(\varphi)$ is shown to model the long-term seasonal effects of air mass on the spectrum, as well as higher frequency spectral shifts due to cloud cover and other atmospheric variables.

4.8. Average photon energy, depth of an absorption band function

The combined APE (φ) and water absorption band depth (ϵ) function builds on $f(\varphi)$ through the inclusion of an additional index [123]. The additional index, ϵ , reduces uncertainty in the original APE SCF,

Table 9

Poly2D model coefficients for each DUT to four significant figures. R^2 values for the corresponding surface fitting functions are included to three significant figures.

DUT	R^2	Model coefficients					
		d_0	d_1	d_2	d_3	d_4	d_5
aSi-T	0.907	-21.94	22.62	-0.01393	-5.521	1.7341E-4	0.003860
CdTe	0.584	-0.5313	0.7208	0.02232	0.05321	1.629E-4	-0.01445
mSi	0.194	-0.3998	1.101	0.03366	-0.1837	1.493E-4	-0.02046

in particular between 1.85 eV and 1.90 eV where similar values of φ yield disproportionately variant values of I_{scn} .

The model is derived and validated using a publicly available empirical dataset developed by the National Renewable Energy Laboratory (NREL) [159]. Two out of every three measurements of the full-year dataset sequenced in chronological order are extracted to build the model development dataset, while the remaining one in three measurements are used to build the validation dataset. This splicing technique ensures data from across the whole year are present in both datasets, thus increasing their representability of the full climatic conditions present through the different seasons of the year. ϵ values for four different wavebands are tested and ranked for each of the three PV technologies investigated. The three PV technologies tested cover a range of semiconductor and construction types, including multijunction (triple junction amorphous silicon), single-junction thin film (Cadmium Telluride), and single-junction crystalline (multicrystalline).

The original publication of $f(\varphi, \epsilon)$ shows that different functional forms may be optimal for different PV technology types and spectral bands of ϵ used in the model. However, a simplified model that strikes a compromise between accuracy and model complexity is derived using a two-dimensional polynomial function of the form:

$$I_{scn} = d_0 + d_1 \cdot \varphi + d_2 \cdot \epsilon + d_3 \cdot \varphi^2 + d_4 \cdot \epsilon^2 + d_5 \cdot \varphi \cdot \epsilon \quad (28)$$

The $f(\varphi, \epsilon)$ model coefficients, d_{0-5} , for the three PV devices investigated in Ref. [123] are summarised in Table 9. The reported R^2 values for each fitting function are also included in the table (see Table 10).

5. Comparison of SCFs

The previous sections show the high level of diversity in methodologies used to model the spectral influence on PV performance and incorporate this influence into forecasts of PV performance. A summary of these methods is presented in Table 10. In this section, a critical comparison is made between the different methods in terms of their underlying principles, such as which parameters are used to characterise the spectrum, and the differences in their predictive performance.

Starting with $f(AM_a)$, this is the most widely-known SCF and the model coefficients have been published for dozens of PV technologies from a range of manufacturers. The free and ready availability of model coefficients make $f(AM_a)$ an obvious choice for PV performance modellers. The ease with which AM_a can be calculated accurately in the field, without the need for specialist equipment and measurements, also makes deriving new coefficients for novel technologies relatively easy. However, air mass is defined by the path length of direct irradiance only, hence spectral effects linked to diffuse irradiance are omitted. Zanesco and Krenzinger [160] highlight that spectral variations are due not only to air mass, but also atmospheric precipitable water, Angstrom's turbidity coefficient, and aerosol particle size distribution. In environments where only the effects of air mass dominate, $f(AM_a)$ may sufficiently capture the majority of the spectral effects on PV performance [140]. However, Hansen et al. [161] find a systematic error in the air mass function where, at the same air mass, the spectral effects on I_{sc} vary depending on the time of day. Klise et al. [162] attribute uncertainty in Sandia Model PV performance predictions, which use $f(AM_a)$, in different geographic locations, to the effect water vapour content and atmospheric turbidity on the spectral distribution.

With all SCFs, quantification of their uncertainties for comparison is difficult due to the differences in the methodologies and datasets used to derive them. Elements of comparison using standardised methods exist in some publications, typically when new models are compared to existing models. In addition, some indication of relative performance may be gleaned from analysing model fitting statistics in the context of the methods through which they were derived. In their proposal of $f(AM_a, W)$, Lee and Panchula [147] compare linear regression results of outdoor testing data for $f(AM_a)$, $f(W)$, and $f(AM_a, W)$. These results are summarised in Table 11.

The absolute values of fitting statistics and errors in Table 11 are not comparable with those of other publications that process their data differently, for example by using different irradiance filtering conditions. However, the relative performance $f(AM_a)$, $f(W)$, and $f(AM_a, W)$ is a meaningful result from Ref. [147]. The combined AM_a, W model decreases the MAE in two of the three locations studied, namely Cocoa (Florida), and Eugene (Oregon). Considering their relative proximity to the sea, in contrast to the continental location of Golden, increased humidity may result in the marked benefit of including W in the SCF. On the other hand, in Golden, the R^2 for $f(AM_a, W)_{mSi}$ is an order of magnitude greater than for the single-variable SCF, yet the MAE value is still higher. This could suggest that the proposed parameterisation for this model is prone to overfitting. Since a mixture of simulated and measured spectra are used, the model development dataset may not be representative of the model validation dataset [24]. Another possibility is that the model-specific coefficients may in fact have a location-specific element to them [46].

None of the three models discussed thus far consider the effects of clouds on the spectrum, and therefore PV performance. $f(AM_a)$ is only defined in clear sky conditions and previous research has shown that it results in high PV performance prediction uncertainty in cloudy conditions. $f(W)$ and $f(AM_a, W)$ are derived using the Simple Model of the Atmospheric Radiative Transfer of Sunshine (SMARTS), which is well-renowned and widely used model [59,163,164], but SMARTS does not account for the influence of cloud cover on the solar spectrum. Furthermore, the field data used for model validation omit cloud cover through the data filtering conditions of $0.7 \leq K_t \leq 1.0$. In environments where air mass and water vapour are the dominant drivers of spectral changes, the omission of clouds may not be significant. However, the effects of cloud cover and non-negligible in moderate climates such as central Europe [42], as discussed quantitatively in Section 2.2.3.

$f(AM_a, K_t)$ is an alternative method that explicitly includes the effects of cloud cover in the spectral correction. Previous work has shown the benefit of including the K_t variable in spectral corrections, compared with the single-variable air mass approach, due to its ability to improve predictions of I_{scn} values greater than unity for amorphous silicon devices [122]. Despite this SCF now omitting the explicit inclusion of atmospheric precipitable water content effects, the PVSPEC parameterisation of $f(AM_a, K_t)$ (Eq. (25)) is shown to exceed the accuracy of the $f(AM_a, W)$ SCF (Eq. (23)) by reducing the root mean square error (RMSE) for all six of the PV technologies tested, which are listed in Table 5. The results of this comparison are presented in Fig. 8. The absolute error of each model is presented in Fig. 8(a), which shows that the PVSPEC model outperforms the $f(AM_a, W)$ model for all technologies save for CIGS. The fact that the RMSE for the JRC model is still lower despite the slightly higher absolute error suggests that $f(AM_a, W)$ may model M with a larger errors, but less frequently, since the RMSE is weighted by the square of the error hence there is a greater penalty for larger errors. Fig. 8 also show that the PVSPEC model is more accurate than the JRC parameterisation of $f(AM_a, K_t)$ (Eq. (24)), which suggests that the PVSPEC parameterisation is a better choice for the PV technologies studied.

The question of choosing between W and K_t as a secondary variable in the AM_a SCF is addressed by Duck and Fell [165], who show that applying a correction for precipitable water to $f(AM_a, K_t)$ can improve the accuracy with which M values are predicted. However, the scale

Table 10

Summary of the SCFs reviewed in this study.

Refs.	PV index	Spectral index	Type	Model form	Data type	Coefficient availability
[140]	I_{scn}	AM_a	Single proxy	4th polynomial	Field	Multiple
[89]	M	W	Single proxy	Exp1D	SMARTS+TMY simulation, field data	CdTe
[123,147]	M	AM_a, W	Double proxy	Poly2D	SMARTS+TMY simulation, field data	CdTe, mSi, aSi-T
[129]	I_{scn}	AM_a, K_t	Double proxy	NL Poly2D	Field	mSi, CdTe
[25]	UF	AM_a, K_t	Double proxy	n/a	Field	n/a
[23]	M	AM_a, K_t	Double proxy	Power	Field	Multiple
[152,153]	M	AM_a, W, AOD	Triple proxy	4th polynomial+log	SMARTS, measured	Multiple
[65]	I_{scn}	AM_a, R_d	Double proxy	Piecewise	One week field	aSi
[122]	I_{scn}	φ	Single direct	4th polynomial	One year field	aSi
[123]	I_{scn}	φ, ϵ	Double direct	Poly2D	One year field	CdTe, mSi, aSi-T

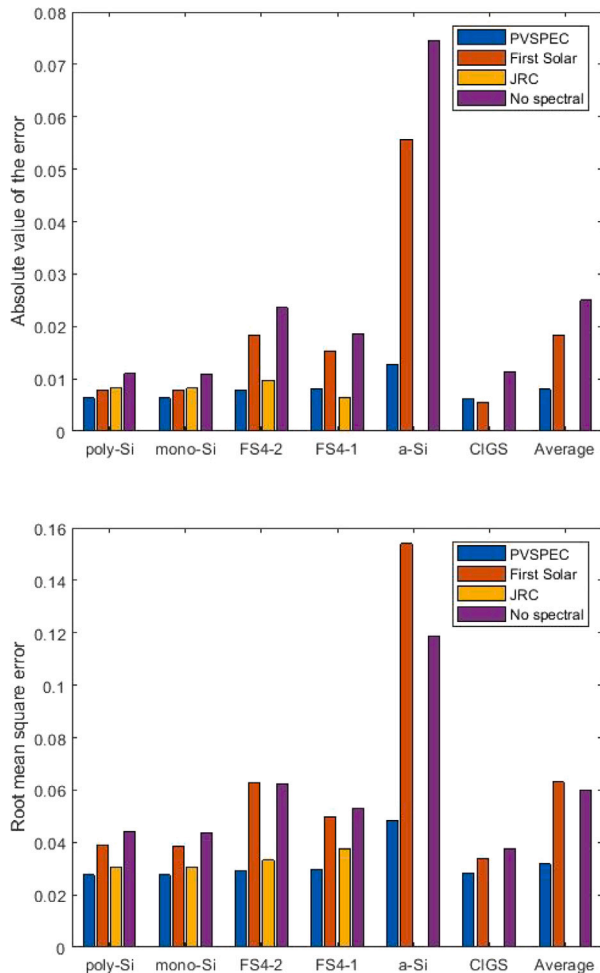
Table 11

Linear regression of outdoor testing data measured for a CdTe and mSi panel in Cocoa (Florida), Eugene (Oregon), and Golden (Colorado) [147].

Site, module	$f(W), f(AM_a)$	$f(AM_a, W)$
Cocoa, CdTe	$MAE_W = 0.0169, R^2 = 0.494$	$MAE = 0.0157, R^2 = 0.705$
Cocoa, mSi	$MAE_{AM_a} = 0.0130, R^2 = 0.428$	$MAE = 0.00749, R^2 = 0.724$
Eugene, CdTe	$MAE_W = 0.0188, R^2 = 0.445$	$MAE = 0.0162, R^2 = 0.598$
Eugene, mSi	$MAE_{AM_a} = 0.00406, R^2 = 0.696$	$MAE = 0.00306, R^2 = 0.817$
Golden, CdTe	$MAE_W = 0.00827, R^2 = 0.712$	$MAE = 0.0150, R^2 = 0.706$
Golden, mSi	$MAE_{AM_a} = 0.00955, R^2 = 0.001$	$MAE = 0.01526, R^2 = 0.356$

Table 12Output energy prediction errors for microcrystalline (mcSi), tandem, and triple-junction modules in 2006 and 2007 using (1) AM_a and K_t , and (2) APE and T_{mod} [169].

Year	Module	K_t and AM_a Error (%)	φ and T_{mod} Error (%)
2006	mSi	0.9	0.6
	Tandem	0.9	1.3
	Triple	0.1	-0.8
2007	mSi	2.4	2.9
	Tandem	2.0	3.1
	Triple	1.8	1.4

**Fig. 8.** Absolute error (a) and root-mean-squared error values (b) for the spectral mismatch prediction of six different PV panels using three different models. Images reproduced with permission from [23].

of this improvement is not quantified in their study. In addition, as the PVSPEC model was released later than the analysis by Duck and

Fell, whether a precipitable water content correction can improve the PVSPEC model has not been investigated.

The $f(AOD, W, AM_a)$ model attempts to maximise the inclusion of spectral proxies in the correction function. The regression statistics for the correlations between modelled and measured M values are greater than 0.8 for all PV devices [152]. Using the same dataset and data processing, Caballero et al. [153] show that the root mean squared errors (RMSE) for predictions of M is lower using $f(AOD, W, AM_a)$ than when using $f(AM_a)$ for all PV devices investigated. However, no comparison is drawn between $f(AOD, W, AM_a)$ and other multi-variable functions, such as $f(AM_a, W)$ and $f(AM_a, K_t)$. Therefore, the increased complexity of including a third variable is not evaluated against any improvement in accuracy, if any is observed. The existing literature has established the geographic dependence of the prevalence of atmospheric precipitable water [166,167], aerosols [168], and other atmospheric components that influence the solar spectrum. Therefore, the worldwide generalisability of proxy-based SCFs is not certain. Duck and Fell [46] suggest that it is this geographic variation in spectral irradiating conditions that leads to the difference they observe between predicted and measured values of M , where the predictions are computed from an SCF derived a site different to that where the measurements are made.

The average photon energy spectral correction function offers a simpler single-variable approach to modelling the spectral influence on PV performance. As it is derived directly from the solar spectrum, the APE is also thought to contain information on all relevant spectrum-influencing variables, including air mass, clearness index, aerosols, atmospheric precipitable water. Takei et al. [169] use the APE and module temperature to predict the energy output of three PV modules and compare these values to the measured output. The APE- T_{mod} prediction errors are compared to those of an air mass and clearness index correlation derived in their study. They conclude that the latter is a better model for output estimation overall given its simplicity and similar accuracy to the APE-based model. Their results are summarised in Table 12. Although Takei et al. are focusing on overall PV system output estimation rather than a specific spectral correction function, their work offers a helpful comparison between the strength of the correlations between PV output and both spectral proxies and a spectrum-derived parameter.

A similar limitation is faced by the work of Williams et al. [116], in which a dimensionless spectral loss factor based on the APE is

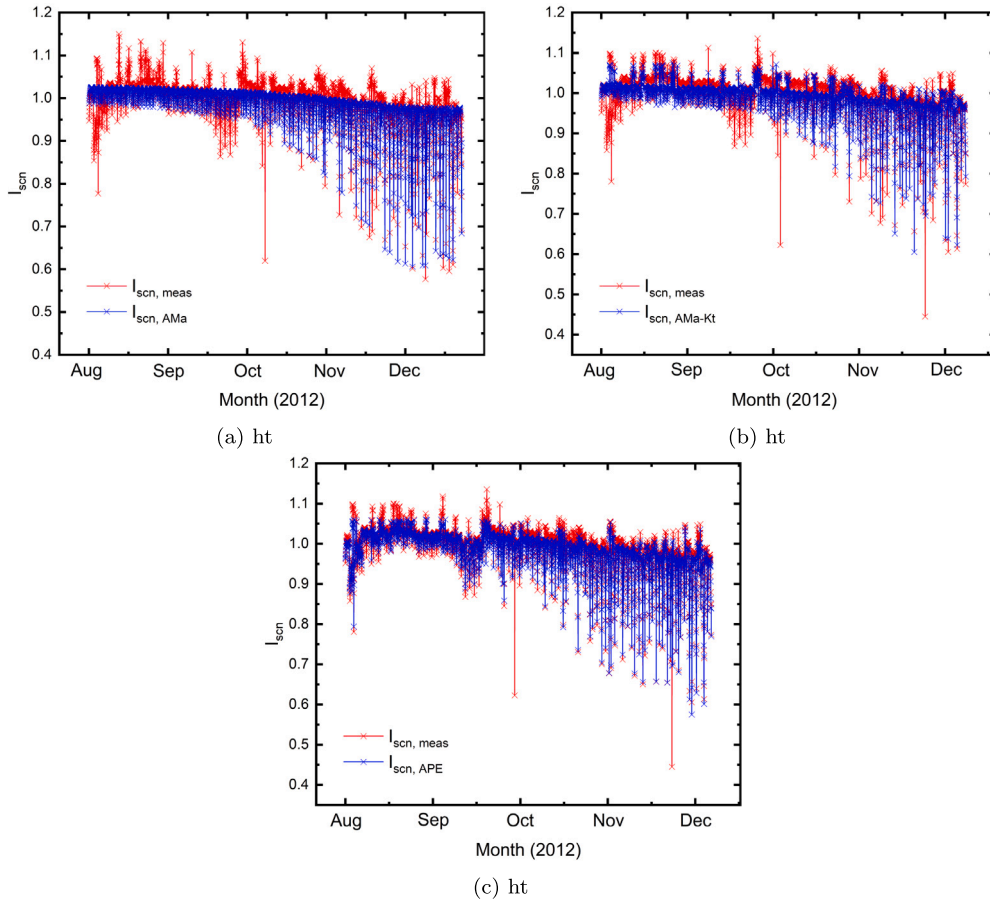


Fig. 9. Time series plots of the predicted I_{scn} values, $I_{scn,calc}$, and those calculated from measured data, $I_{scn,meas}$.

proposed. A single spectral loss factor is reported for each PV device based on their experiment campaign, although it is not possible to identify the values due to the low resolution of the presented bar chart (Fig. 5 in [116]). Besides, their work focuses more on the overall PV performance model, which is reported to be accurate to within 7% of the measured performance. The coefficients of the spectral loss factor model are not published, and the accuracy of the proposed spectral loss factor is not analysed in isolation from the other loss factors that form part of the wider PV performance model. Therefore, it is not possible to compare the performance of their proposed SCF to that of the other SCFs discussed in this section. In addition, only data for a short winter period between October 2002 to April 2003 are used, which may limit the seasonal generalisability of the model. Finally, it may be argued that the time series analysis is somewhat more complex than some of the simpler approaches, for example the polynomial parameterisation of $f(AM_a)$ or even two-variable surface parameterisation of $f(AM_a, W)$.

In work focused more specifically on spectral correction functions, Ref. [122] shows that an SCF based solely on APE can in fact outperform one based on AM_a and K_t by a significant margin [122]. $f(\varphi)$ is shown to yield a reduction in the mean absolute error of almost 40% when compared to $f(AM_a, K_t)$, and almost 50% when compared to $f(AM_a)$. The results from Ref. [122] are quantified through linear regression analysis and a time series comparison of the measured and predicted I_{scn} values for each SCF, the latter of which is shown in Fig. 9.

$f(AM_a)$ is shown to be capable of modelling only the long-term seasonal shift in I_{scn} as the solar elevation angle changes throughout the year. $f(AM_a, K_t)$ models the long-term shift as well, but the K_t parameter enables the correction function to predict higher frequency gains in I_{scn} above unity for an amorphous silicon PV device. This is expected given that cloud-induced scattering of solar irradiation

Table 13

Fit statistics for the AM_a , AM_a-K_t , and APE spectral correction functions absolute predictive accuracy test [122].

Correlation	Statistic		
	r	R^2	MAE
$I_{scn,meas}(AM_a)$	0.874	0.765	0.024
$I_{scn,meas}(AM_a, K_t)$	0.898	0.806	0.022
$I_{scn,meas}(APE)$	0.969	0.940	0.014

shifts the solar spectrum to shorter wavelength (higher APE), which increases the proportion of total irradiance falling within the spectral response window of an amorphous silicon device. The $f(\varphi)$ SCF is shown to predict high frequency gains and losses in I_{scn} as well as the long-term seasonal trend. Compared with the combination of air mass and clearness index, it is clear that the APE parameter contains more information on the environmental variables that affect the solar spectrum, and thus the performance of PV devices. A comparison of the mean absolute errors of the I_{scn} values as predicted by the three SCFs — $f(AM_a)$, $f(AM_a, K_t)$, and $f(\varphi)$ — is presented in Table 13. This table also includes the Pearson r and R^2 values calculated from the linear regression of the predicted and calculated I_{scn} values for each SCF.

One shortfall of the aforementioned study is that the proposed APE spectral correction methodology is only demonstrated using an amorphous silicon device. Further work by the same authors shows that the uncertainty in APE values used to characterise spectral distributions, discussed in Section 2.2.6, increases the uncertainty in $f(\varphi)$ when applied to other PV devices such as CdTe and mSi [123]. One approach to more accurate spectral characterisation using the APE parameter is to combine it with a value for the depth of a water absorption

Table 14

Annual MAE values for the predictions made by $f(\varphi)$, $f(AM_a, W)$, and $f(\varphi, \varepsilon)$ for mSi, CdTe, and aSi-T PV devices [123].

DUT	MAE		
	$f(\varphi)$	$f(AM_a, W)$	$f(\varphi, \varepsilon)$
mSi	0.0112	0.0250	0.0102
CdTe	0.0152	0.0187	0.0149
aSi-T	0.0149	0.0224	0.0134

band or atmospheric window, ε [121]. By adopting this approach, the authors propose a new spectral correction function based on both φ and ε . Several candidate wavebands for *varepsilon* are analysed and it is determined that the 650–670 nm band is the optimal choice.

Compared with the single-variable $f(\varphi)$ SCF, $f(\varphi, \varepsilon)$ reduces the MAE of I_{scn} predictions for aSi-T and mSi by 10%, and by 2% for CdTe. The results are also compared to predictions made by the $f(AM_a, W)$, and improvements in the MAE of 60%, 20%, and 40% are achieved by the new model for the mSi, CdTe, and aSi-T modules, respectively [123]. The MAE values reported in the study are summarised in Table 14.

Both $f(\varphi, \varepsilon)$ and $f(AM_a, W)$ attempt to include the effects of atmospheric water vapour, albeit through using different parameters. The main difference between the models is that one is based on parameters derived directly from the spectrum while the other uses proxy variables to represent the spectrum. The results suggest that the direct parameters offer greater accuracy than the proxy variables. However, any improvements in accuracy achieved in this way need to be balanced against the availability of spectral irradiance data at the test location.

6. Discussion and recommendations for further work

Although there seems to be a consensus on the choice of methods to characterise the variation in PV performance due to variation in the spectrum, with **most studies adopting the mismatch factor or normalised short-circuit current**, with a few variants thereof, there is great debate over the choice of spectral characterisation method. One of the likely causes of this range of methods being promoted is the lack of standardised testing for the different methods, particularly in different locations, climates, and for different PV systems. Although many individual studies offer comparisons between different SCFs, the methodologies between studies are disparate, yielding intercomparisons of a wider range of SCFs for application in and to a wider range of climates and PV technologies practically impossible. However, this review finds that, overall, the available literature does seem to suggest that the direct spectral characterisation methods are likely to be more accurate than the proxy variable methods in most situations. However, spectral irradiance measurements are not currently widely utilised at PV plants and test sites. **Therefore, the most accurate model may not be the most suitable for the end user.** For example, in the absence of spectral irradiance data, be it simulated or measured, $f(\varphi)$ would be inappropriate. In another scenario where the test site typically experiences cloudless skies but has high levels of atmospheric aerosol pollution, $f(AM_a, AOD, W)$ might be a better choice than $f(AM_a, K_t)$. Based on the analysis in this review, a decision-making framework for PV performance modellers is proposed. Fig. 10 is a butterfly diagram demonstrating one possible construction of this framework that PV performance modellers could use to determine which SCF to use based on their PV system specifications, local climate, data availability, requirements for accuracy, etc.

Fig. 10 is based on the results from this review alone, so does not include details that would be necessary in a more rigorous framework. For example, no distinction is made different types of data, namely simulated and measured, as a comparison between the resultant model accuracies using these different types of data have not been analysed in the literature. Another example is the distinction between $f(AMCDF)$

and $f(AM_a, K_t)$. Without a standardised comparison between the different SCFs, it is difficult for a PV performance modeller to decide which to use based on their use case scenario. If the research gaps highlighted in this section are addressed, a more detailed and precise framework could be developed.

In order to support the construction of a more accurate decision-making framework and move the field of spectral correction modelling forward, this review highlights two main areas of required research development. These two areas are as follows:

1. The need for a standardised comparison of the different SCFs currently available, for different PV modules, climate regions, and use case scenarios.
2. Further development of SCFs based on parameters derived directly from the spectrum, in particular through validation when simulated model input parameters are used.

The first suggestion for future work arises from the fact that although a plethora of different methods exist to model the spectral influence on PV performance, these methods have been derived and validated using entirely different methodologies. The key differences in the methodologies include the data acquisition methods, data processing and filtering conditions, the type of PV devices investigated, the climate region under which the investigation is undertaken, and the seasons of the year during which the investigations are undertaken. These differences render the results presented for different models incomparable.

Therefore, standardised measurement campaign and data analysis protocols should be deployed in a range of different climate regions for a set of the same, or at least comparable, PV modules. The immediate benefit of such a campaign would be the possibility to compare reliably and accurately the performance of different spectral correction functions. Not only would these results indicate the relative accuracies of the different models, but also enable the determination of their best use case scenarios.

The second recommendation for future work is on the subject of developing specific spectral correction models. The models based on parameters derived directly from the spectrum, such as APE and ε , rather than proxy variables, such as AM_a and W , are more accurate. Despite their lower accuracy, proxy variables can easily be calculated based on time of day and other routine and cheap measurements of variables such as temperature and humidity. While spectrometers are falling in price [170], they are still relatively expensive in comparison with other meteorological sensors. Spectral irradiance data are not typically measured at PV test sites and power plants either. Therefore, the spectrum-based SCFs suffer a limitation in terms of the availability of their required input parameters. Nevertheless, many spectral simulation models exist [59,111,171,172] and, in more recent years, have been developed to include the effects of more complex weather conditions such as cloud cover [173]. The benefit of spectral-based models is clear, but these models that have used empirical spectral irradiance data should be validated with simulated spectral irradiance data. If such models can maintain their high level of accuracy with simulated input spectra, their usability and application potential would be enhanced greatly.

If the two research gaps highlighted in this section are met, another review of published SCFs, using standardised results and both measured and simulated input parameters, would be useful. The results of such a review could guide PV performance modellers in their selection of models based on PV system specifications, climate and geography, forecasting frequency required, availability of data and whether it is measured or simulated, and model complexity.

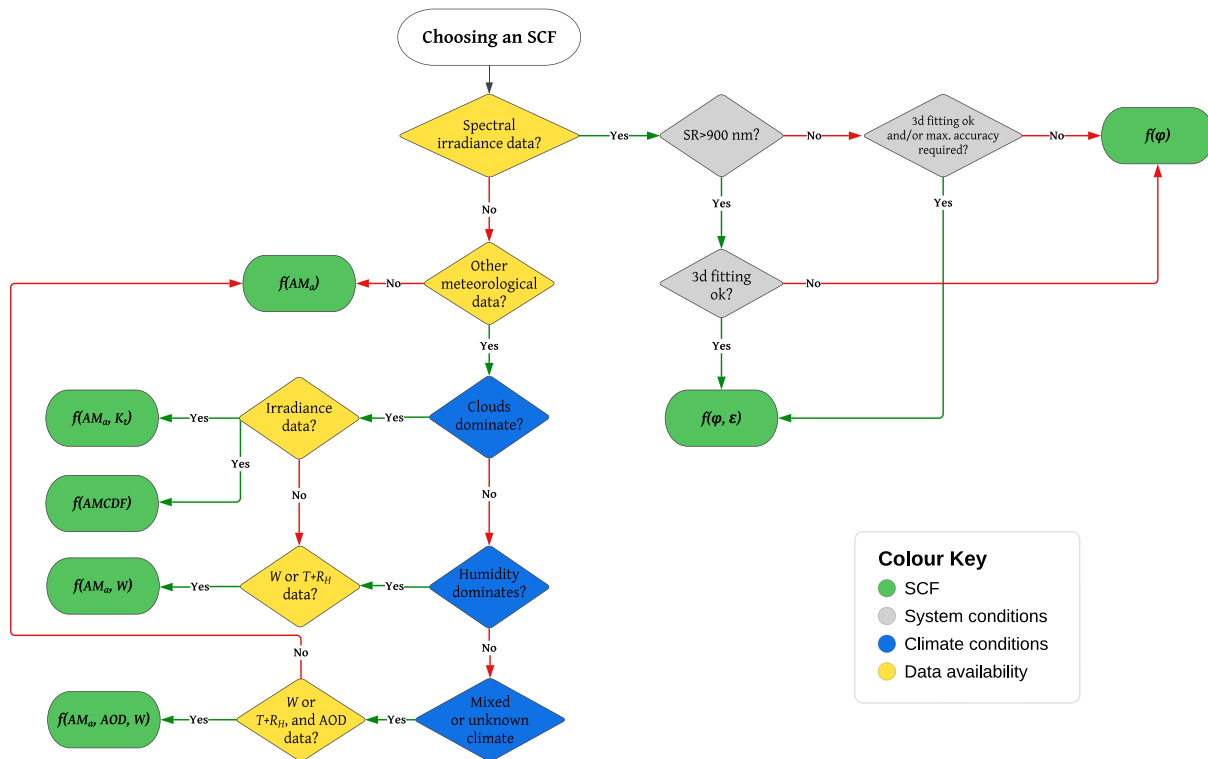


Fig. 10. Butterfly chart illustrating a possible review and recommendation structure for SCFs based on different users' requirements. These requirements include the specifications of system being investigation (computational capacity for analysis, types of devices being tested), climate under which the system is operating, and the availability of data. R_H is relative humidity, T is temperature, SR is spectral response, max. is maximum, and all other symbols take their usual definitions as defined previously.

7. Conclusion

Models for the solar spectral influence on PV performance are a crucial component of PV performance models. These models apply a correction to a reference performance value to account for the influence of the spectrum, and are hence known as spectral correction functions (SCFs). Currently, a wide range of different SCFs exist. Some of these models use input parameters derived directly from the spectrum, while others use proxy variables to represent the solar spectrum.

This review has shown that the proxy variable SCFs are a mature type of model and offer a relatively easy way to account for the spectral influence on PV performance. However, the applications may be more case-specific than currently evident in the literature, due to the limited range of information the dependent variables contain on the shape of the solar spectrum. Other models whose input parameters are derived directly from the solar spectrum, such as the average photon energy and the depth of water absorption bands or atmospheric windows, offer a more widely applicable solution to spectral correction modelling. However, the simulation of such spectral parameters is more complex than that of the proxy variable parameters. The published spectral parameter-based SCFs have not been validated using simulated inputs.

A standardised measurement campaign is necessary to compare the relative accuracies of the different SCFs and determine when and where each model is the optimal choice. Key areas for investigation through this kind of standardised campaign are the performance of different models in specific climate conditions and for different PV devices. In addition, a comparison between model performance based on the use of both measured and simulated data is necessary, in particular for spectrum-based SCFs such as $f(\phi)$ and $f(\phi, \epsilon)$.

A decision-making framework for choosing the optimal SCF based on different use case scenarios has been hypothesised based on the analysis in this review. This framework could be validated and developed further based on the results of a standardised investigation as described in this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in the manuscript are available from the cited literature and simulation software.

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